EKRA ISLAM OHI

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EDUCATION

Green University of Bangladesh | B.Sc. in Computer Science and Engineering — 2023 - 2027 (Expected)

Dhaka, Bangladesh

- **Coursework:** Data Structures & Algorithms, Machine Learning & Pattern Recognition, Operating Systems, Database Management Systems, Object-Oriented Programming (Java/C++), Computer Networks, Software Engineering.

RESEARCH MANUSCRIPTS & WORKING PAPERS

Explainable Deep Transfer Learning for Rice Leaf Disease Classification UNDER PEER REVIEW — 2025

Authors: Ekra Islam Ohi, et al. | Conference Manuscript Submission (Faculty Supervised)

- Trained ResNet-18 architectures across 8 botanical condition categories, evaluating accuracy and loss convergence.
- Implemented Deep Convolutional GAN (DCGAN) synthetic patch generation to address severe agricultural class imbalance.
- Generated Grad-CAM saliency heatmaps to provide transparent, interpretable visual evidence for disease localization.

SELECTED RESEARCH SYSTEMS & SOFTWARE PROJECTS

CitationGuard AI | Evidence-Aware Citation Stance Verification — Python, PyTorch, PubMedBERT

- Fine-tuned PubMedBERT to verify scientific claims against biomedical literature into 3 classes (Supported, Contradicted, NEI).
- Enforced strict context isolation protocols to test attribution fidelity without relying on ungrounded model memory.

Bangla Document Intelligence | Bilingual OCR & In-Memory Search — Python, EasyOCR, PyMuPDF, Docker

- Constructed an OCR extraction pipeline converting multi-page PDFs to text with bilingual Bengali and English support.
- Built an in-memory inverted text index providing instant keyword search across digitized documents.

Butterfly Species Classification & XAI | Vision & Explainability — PyTorch, ResNet-18, Streamlit

- Trained deep transfer learning classifier on 8 butterfly species with custom data augmentation pipelines.
- Integrated native PyTorch forward/backward hooks to render real-time Grad-CAM explainability heatmaps in web UI.

Momin Agro | Dairy & Biogas Management PWA — React, TypeScript, Supabase, PostgreSQL

- Engineered operational management software with PostgreSQL Row Level Security (RLS) for cattle logs and milk yield records.

UniPlanner | Academic Course & GPA Forecasting Tool — TypeScript, React, PWA, Service Worker

- Built an offline-first progressive web app with Service Worker caching supporting Bangladesh UGC 4.0 and US 4.0 scales.

TECHNICAL COMPETENCIES

Programming Languages:	Python, C, C++, TypeScript, JavaScript, SQL
AI / Machine Learning:	PyTorch, Hugging Face Transformers, Scikit-Learn, OpenCV, Grad-CAM, DCGAN, BERT
Tools & Engineering:	Docker, Git & GitHub, Linux/Bash, React, Node.js, Fastify, PostgreSQL, Supabase, Vite, PWA
Research Methodologies:	Context Isolation, Stance Verification, Document OCR Pipelines, Synthetic Augmentation, XAI Saliency

RESEARCH INTERESTS & ACADEMIC DIRECTION

Focusing on Machine Learning Reliability, Natural Language Processing, and Interpretable Computer Vision. Preparing for international graduate studies (M.S./Ph.D.) and research-driven software engineering roles dedicated to dependable, verifiable intelligent systems.